

Problem Set 4: Functions and Modules

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1 Exercise 1: Patient Data Module

1. Create a Python module called

`patient_data.py`

The module will organize a block of fictional patient records. Each patient should be represented by a tuple containing

```
(name, ssn, age, dob,
 height, weight, gender)
```

Use the following data:

```
patients = [
    ("Alan Reed", "000-00-0001", 74,
     "04/12/1952", 70, 218, "M"),

    ("Maria Lopez", "000-00-0002", 76,
     "09/03/1950", 64, 155, "F"),

    ("Robert Chen", "000-00-0003", 68,
     "01/19/1958", 72, 225, "M"),

    ("James Hill", "000-00-0004", 82,
     "11/28/1943", 69, 210, "M"),

    ("Susan Patel", "000-00-0005", 72,
     "06/14/1954", 65, 205, "F"),

    ("David King", "000-00-0006", 71,
     "02/07/1955", 71, 195, "M"),

    ("Henry Cole", "000-00-0007", 79,
     "08/21/1947", 68, 242, "M"),

    ("Linda Grant", "000-00-0008", 66,
     "12/02/1959", 63, 180, "F")
]
```

The SSNs above are fictional identifiers created only for this programming exercise.

2 Exercise 2: Module Functions

1. Inside `patient_data.py`, define the following functions.

- A function that returns the complete patient list:

```
def get_patients():
    """Return all patient records."""
    return patients
```

- A function called

```
def screening_rule(patient):
```

that returns `True` only when all three conditions are met:

```
gender = male,    age > 70,    weight > 200.
```

Use the tuple indices and Boolean operator `and`. For example,

```
age = patient[2]
weight = patient[5]
gender = patient[6]
```

The function must **return** a Boolean value rather than printing the result.

- Define a third function

```
def display_patient(patient):
```

that prints the patient's information in a readable format.

For example,

```
Name: Alan Reed
SSN: 000-00-0001
Age: 74
DoB: 04/12/1952
Height: 70 in
Weight: 218 lb
Gender: M
```

Include a short docstring inside each function.

3 Exercise 3: Screening Program

1. Create a second Python file called

`heart_screen.py`

Import your module using

```
import patient_data
```

Retrieve the patient records:

```
patients = patient_data.get_patients()
```

Then use a `for` loop to examine each patient:

```
for patient in patients:
```

```
    if patient_data.screening_rule(
        patient):
```

```
        patient_data.display_patient(
            patient)
```

```
        print("")
```

Your program should display **only** patients who satisfy all three screening conditions.

- Do not repeat the screening logic in `heart_screen.py`. The decision must be made by the function inside the module.
- Do not manually type the names of qualifying patients.
- Verify that changing one patient's age, weight, or gender in `patient_data.py` can change whether that patient appears in the final output.
- Add the heading

Patients Matching Screening Rule

before displaying the selected records.

4 Check Your Program

1. Your submission should contain exactly two Python files:

```
patient_data.py
heart_screen.py
```

The module should store the patient data and define reusable functions. The main program should import the module and call those functions.

For this programming exercise, the three-condition rule is only a software-filtering example and should not be interpreted as an actual medical assessment of heart-disease risk.